People's Democratic Republic of Algeria
2022-2023 Ministry of Higher Education and Scientific Research
National Higher School of Mathematics

NATIONAL HIGHER SCHOOL OF MATHEMATICS
المدرسة الوطنية العليا في الرياضيات

Final exam (2^h30)

Name :	Groupe:

Exercice 1. Among these assertions, specify which ones are true and which ones are false.

	True	**False**
1-The median is always higher than the mean.		
2- If $P(A \cap B) \geq 0.10$ then $P(A) \geq 0.10$.		
3- If $P(A\|B) = 0.5$ and $P(B) = 0.5$, then A and B are necessarily independent.		
4- If a value is higher than the mean then it is an outlier.		
5- $P(A) - P(B) \neq P(A \cap B^c)$.		
6- Multiplying all values by 2, will multiply the STD by 2.		
7- Removing an outlier that is negative, will most likely decrease the STD.		
8- If A and B are mutually exclusive events, then $P(A\|B) = P(A)$.		
9- The mean is equal to 0 for a symmetric histogram.		
10- $E[X + 2Y] = E[X] + 2E[Y]$.		
11- The probability is always non negative.		
12- The STD of a sample is always less than 1.		
13- Two r.v X and Y are identically distributed if $\exists x \in \mathbb{R}, F_X(x) = F_Y(x)$.		
14- The median is always higher than the mean.		
15- If $A \cap B \neq \emptyset$ and the events B and C are disjoint. then $A \cap B \cap C^c = \emptyset$		
16- If $P(A\|B) = 0.6$ and $P(B) = 0.2$ then $P(A \cap B) = 0.12$.		
17- $P(A_1 \cup A_2 \cup ...) \leq P(A_1) + P(A_2) + ...$		
18- The Coefficient of determination R^2 is a unitless number between -1 and 1.		
19- $\forall a, b, c \in \mathbb{R}, Var(aX + bY + c) = a^2Var(X) + b^2Var(Y) + 2abCov(X;Y)$.		
20- We can learn the IQR by looking at the boxplot.		

Exercice 2. For a sample of 15 students we observe the Time (in minutes, X) usually spent using Facebook per day, and the final grade in the Statistics exam (Y):

x_i	0	11	17	16	22	17	25	30	27	27	31	35	30	45	60
y_i	25	22	28	30	22	27	26	21	27	28	23	29	30	24	27

1. Compute mean, median and quartiles for both X and Y

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2. Find the standard deviation and the coefficient of variation for both X and Y. Which variable is more scattered?

3. Plot the box-plot for both X and Y and comment them.

Exercice 3. Since clean-air regulations have dictated the use of unleaded gasoline, the supply of leaded gas has diminished. The table below was given

Year	1981	1988	1992	1996	2000
Quantity ($\times 10^3 L$)	150	124	104	76	50

1. Draw the scatter plot between the two variables and measure the correlation between them.

2. Create a linear model to predict the quantity that would be available in 2005.

3. Create a model that would estimate the year when leaded gasoline will first become unavailable then draw this line on the above scatter plot.

4. Give the value of R^2 for the line of the 3^{rd} question then briefly interpret its value.

Exercice 4. Consider a given population, a certain disease D and a certain symptom S. We know that 85% of the population members who have the disease D do have the symptom S, while the remaining 15% do not. Suppose that 95% of those who do not have the disease D do not have the symptom S, while 5% do have the symptom S (due to some other cause).
Suppose that 10% of the given population have the disease.

1. What is the probability that a random person chosen from the population has the symptom?

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2. Given that a random person was sampled, and he has the symptom, what is the probability that he or she has disease?

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3. What is the probability that a random person does not have the disease and does not have the symptom?

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